

Project Overview:

Project includes 5 conveyor belts, for moving French fries from line A to storage bin. Belts will start in reverse order (5,4,3,2,1) and stop in forward order (1,2,3,4,5)

Sequence of Operations:

1. When operator starts the system, belt 5 starts first
2. After 1 second time delay, belt 4 starts
3. After 1 second time delay, belt 3 starts
4. After 1 second time delay, belt 2 starts
5. After 1 second time delay, belt 1 starts
6. When operator stops the system, belt 1 stops after 1 second delay
7. After 1 second time delay, belt 2 stops
8. After 1 second time delay, belt 3 stops
9. After 1 second time delay, belt 4 stops
10. After 1 second time delay, belt 5 stops

IO List:

Tag Name	Description	PLC Address	Tag Type
B1_Running	Belt 1 running	Local:1:I.Data.0	DI
B1_Fault	Belt 1 faulted	Local:1:I.Data.1	DI
B1_Run_CMD	Belt 1 fault command	Local:2:O.Data.0	DO
B2_Running	Belt 2 running	Local:1:I.Data.2	DI
B2_Fault	Belt 2 faulted	Local:1:I.Data.3	DI
B2_Run_CMD	Belt 2 fault command	Local:2:O.Data.0	DO
B3_Running	Belt 3 running	Local:1:I.Data.4	DI
B3_Fault	Belt 3 faulted	Local:1:I.Data.5	DI
B3_Run_CMD	Belt 3 fault command	Local:2:O.Data.0	DO
B4_Running	Belt 4 running	Local:1:I.Data.6	DI
B4_Fault	Belt 4 faulted	Local:1:I.Data.7	DI
B4_Run_CMD	Belt 4 fault command	Local:2:O.Data.0	DO
B5_Running	Belt 5 running	Local:1:I.Data.8	DI
B5_Fault	Belt 5 faulted	Local:1:I.Data.9	DI
B5_Run_CMD	Belt 5 fault command	Local:2:O.Data.0	DO

Alarm List:

Tag Name	Description	Alarm Type	Alarm Condition
B1_Fault_ALM	Belt 1 fault alarm	Digital	ON
B1_FTS_ALM	Belt 1 fail to start alarm	Digital	ON
B2_Fault_ALM	Belt 2 fault alarm	Digital	ON
B2_FTS_ALM	Belt 2 fail to start alarm	Digital	ON
B3_Fault_ALM	Belt 3 fault alarm	Digital	ON
B3_FTS_ALM	Belt 3 fail to start alarm	Digital	ON
B4_Fault_ALM	Belt 4 fault alarm	Digital	ON
B4_FTS_ALM	Belt 4 fail to start alarm	Digital	ON
B5_Fault_ALM	Belt 5 fault alarm	Digital	ON
B5_FTS_ALM	Belt 5 fail to start alarm	Digital	ON